PATHOPHYSIOLOGY — NURSING

The Cell — Part 1

NRSG 224 — Class 1. Exam-detail notes from the recorded lecture slides and her study-guide handout.

Same format as the immune notes. Black is the lecture and the filled handout. **Red is a blank she left**, filled from the class deck and the later pages of the same handout. Order follows the recorded slides so you can follow the video.

1. How she writes the test

Class slides 2–4, before the cell content. She writes questions on things students mix up: similar words, diseases with more than one type or an opposite, and any time there is more than one kind of something.

- Wrong options are often the opposite of the right answer, something from another disease in the same system, or a word she used in that unit that does not answer the question.
- When you study: check the vocab for similar terms, name the diseases that are easy to mix up, and say how they are alike and how they are unique.

2. The cell and its parts

The cell is the basic unit of a living organism. If a part fails, the nursing problem is the job that part can no longer do.

Nucleus — control center

Holds DNA, the instructions for essential proteins and the genetic information passed to the next generation. Makes RNA, which carries those instructions to the cytoplasm, where protein synthesis happens, and helps make the protein.

Blank: problems with the nucleus are bad instructions. Defective proteins, or genetic disorders, because the recipe is wrong before the ribosome ever starts.

Ribosomes

Sites of protein synthesis.

Blank, and the practice item: impaired ribosomes mean synthesis of defective proteins, or decreased protein synthesis. Not the energy answer. That one is mitochondria.

Endoplasmic reticulum

A tube system that transports substances through the cell and houses ribosomes and the enzyme systems for the cell's metabolic work.

Blank, and the practice item: if the ER fails, substances may not be transported appropriately within the cell. Not endocytosis. That is the membrane.

Mitochondria

Turn food energy into ATP, which powers cell activities. Aerobic metabolism happens here.

Blank: the cell cannot generate enough energy to maintain function. Practice slide uses that line as the wrong answer for ribosomes. It belongs here.

Lysosomes and peroxisomes

Lysosomes are bags of enzymes that digest worn-out cell parts and foreign material, such as bacteria, that got into the cell. Peroxisomes are similar. They hold enzymes that detoxify substances and help control free radicals, highly unstable compounds that can damage cytoplasm.

Blank: worn-out parts and foreign material accumulate. Her recorded slide names lysosomal storage disease, example Tay-Sachs. The next named slide is I-cell disease, which sits with the packaging problem: enzymes never get delivered into the lysosome.

Golgi complex

Modifies what the ribosomes made, packages it into secretory granules or vesicles, and directs it where it is supposed to go.

Blank: substances are made but not modified, packaged, or sent to the right place. Hormones and lysosomal enzymes both depend on this step.

Cell membrane

- Holds the cell together. Barrier against foreign organisms and toxic molecules.
- Proteins that move substances in and out: carrier proteins, ion channels.
- Receptors for hormones and other active substances.
- Antigens that label the cell self or nonself.
- Maintains and transmits membrane potentials.

Blank: a membrane problem can be a transport failure, a receptor failure, a self-label failure, or a potential that will not fire. Her recorded slide names myasthenia gravis here: antibody against the receptor, so the message does not land.

TEST TIP. Match the broken part to its one job. Ribosome = make protein. ER = move it inside the cell. Golgi = package and address it. Mitochondria = ATP. Lysosome = digest debris. Membrane = in, out, receptors, potential.

3. Making energy

Cells use oxygen and food to make the energy they need. Energy is stored as ATP. When the cell needs it, ATP is broken down and energy is released. Two pathways.

	Aerobic	Anaerobic
Oxygen	Requires oxygen.	No oxygen.
Where	Mitochondria.	Cytoplasm.
Point	The efficient path.	Less efficient. Matters when oxygen delivery is low.

Blank on the metabolism row: altered energy production. Predisposing factors she listed: less oxygen to the cell, less food or fuel, or metabolic enzymes that are missing or not working.

4. Movement across the membrane

Required to keep the cell working. Three ways.

Passive — no energy

Follows a chemical or electrical gradient, high to low, trying to even out.

- Diffusion: particles move down the concentration gradient.
- Osmosis: water moves down its gradient through a semi-permeable membrane until solute concentration is equal on both sides.

Her deciding factor, on the slide: cell membrane permeability. Goal: equilibrium. Usually more K+ inside, more Na+ outside. K+ can diffuse out. Na+ cannot get in. Water moves instead. That water movement is osmosis.

Hypernatremia slide: more water leaves, the cell shrinks.

Active transport — energy required

Used when a substance has to move against the concentration gradient. The sodium-potassium pump is the one she builds the whole potential on.

Practice item: the purpose of active transport is to let substances move in and out against the concentration gradient. Not enzyme activation, not free-radical protection, not the signal itself.

Endocytosis and exocytosis

- Endocytosis: the cell engulfs material, forms a vesicle, and brings it in. Pinocytosis is fluid and dissolved particles. Phagocytosis is microorganisms, damaged cells, and other particles, which are then killed or broken down.
- Exocytosis is the reverse. A secretory granule fuses with the membrane and opens outside. How the cell releases hormones it made, and how it dumps debris.

DO NOT MIX UP. Diffusion moves particles. Osmosis moves water. Which one happens depends on whether the membrane lets that particle through. Active transport is the only one that spends ATP, and the only one that goes uphill.

5. How cells talk, and the membrane potential

Two methods

- Chemical messengers bind receptor proteins on other cells and trigger the effect.
- Gap junctions, in nerve and muscle, are channels that let small ions and molecules pass straight from cell to cell, so an action potential spreads fast through that tissue.

Blank: the messenger or the receptor is missing or not working. Predisposing factors she listed: autoimmune disease, congenital defects, genetic predisposition, degeneration from age or toxins. Myasthenia gravis is the receptor version.

Resting membrane potential

All body cells are electrically polarized. At rest the inside is more negative than the outside. A change in that polarization can carry a message and trigger work, such as muscle contraction or a nerve impulse.

- The sodium-potassium pump keeps sodium higher outside and potassium higher inside.
- The resting membrane is more permeable to potassium than sodium, so potassium diffuses out, down its gradient.
- What is left inside is mostly negatively charged ions. That is the resting potential.

Action potential

- Depolarization: a stimulus makes the membrane more permeable to sodium. Sodium rushes in, down its gradient. The inside is no longer more negative. That is the action potential. Current-carrying ions pass through gap junctions, and the impulse spreads through muscle or nerve.
- Repolarization, two steps on her slide. First the membrane goes back to its original permeability, less open to sodium and more open to potassium. Then the sodium-potassium pump moves sodium back out and potassium back in, and the cell is ready to start again.

TEST TIP. Short answer she asked: why is oxygen necessary for a cell to repolarize? The pump that restores sodium and potassium is active transport. Active transport needs ATP. The efficient ATP path is aerobic, in the mitochondria, and that needs oxygen. No oxygen, no pump, the cell cannot get back to resting potential.

Blank on this row: changes in membrane permeability. Predisposing factors: electrolyte imbalances, and hypoxia or impaired metabolism so there is not enough energy for the pump.

6. Cell injury

Free radicals

Highly unstable atoms that react with key molecules. They damage membranes and proteins, inactivate enzymes, and damage DNA. Byproducts of normal work, including oxidative reactions in metabolism and inflammation. Outside sources she named: tobacco smoke, pollutants, pesticides, radiation.

Hypoxia and ATP depletion

Power failure in the cell. Her flowchart: inadequate oxygen goes three ways, and all of them can end in cell death.

- Inadequate energy, then metabolic failure.
- Switch to anaerobic metabolism. Lactic acid. Lower pH. Metabolic acidosis. Membranes get damaged, and intracellular enzymes leak out. The enzymes she named on the slide: CK (creatine kinase), LD (lactate dehydrogenase), troponin.
- Lysosomal membranes can break too. Lysosomal enzymes leak into the cytoplasm and the cell digests itself.
- Sodium-potassium pump fails. More sodium and water enter. The cell swells. Opposite of the hypernatremia shrink.
- Other energy-dependent transport fails. Membrane permeability rises. Proteins and RNA leak out. More sodium and water in. More calcium in, which turns on enzymes that damage the cell.

Predisposing factors she listed

- Physical: mechanical forces, temperature extremes, electricity.
- Radiation.
- Chemicals: drugs, environmental toxins such as lead, tobacco smoke.
- Biologic: microorganisms, parasites.
- Nutrition, both ways. Excess, her example obesity leading to atherosclerosis. Deficiency, her example iron-deficiency anemia.
- Others: genetic derangements, autoimmune disorders.

7. What the cell does about it

Three boxes on the slide: cellular adaptation, reversible injury, cell death. The handout fills the adaptation list. These were the nursing-problem lines she actually wrote out.

- Atrophy: the cell goes back to a smaller size and a lower level of function. Oxygen and energy needs fall, so survival is more likely.
- Hypertrophy: bigger cell, more functional parts, so it can meet a higher work demand.
- Hyperplasia: more cells, not bigger cells.
- Metaplasia: one cell type changes to another that can survive the adverse condition. Usually chronic irritation or inflammation.
- Dysplasia: deranged growth. Cells vary in size, shape, and appearance. Usually chronic irritation or inflammation.
- Reversible injury: the cell is hurt and can still recover.
- Cell death / necrosis: the injury is irreversible.

DO NOT MIX UP. Hypertrophy is size. Hyperplasia is number. Metaplasia is a change of type. Dysplasia is deranged type. Atrophy is smaller and quieter. Metaplasia and dysplasia both follow chronic irritation. Dysplasia is the one that looks abnormal.

8. Practice items from the class deck

- Ribosomes impaired: synthesis of defective proteins, or decreased protein synthesis.
- ER impaired: substances may not be transported appropriately within the cell.
- Active transport: move substances against the concentration gradient.
- Oxygen and repolarization: the sodium-potassium pump needs ATP, and aerobic ATP needs oxygen.
- Accumulation of worn-out parts is the lysosome answer, not the ribosome answer. Unable to make enough energy is the mitochondria answer.

9. Rapid-fire

- Nucleus = recipe. Ribosome = make the protein. ER = transport inside. Golgi = package and address. Mitochondria = ATP. Lysosome = digest. Peroxisome = detox and free radicals. Membrane = barrier, transport, receptors, self label, potential.
- Aerobic = mitochondria, with oxygen. Anaerobic = cytoplasm, no oxygen, lactic acid, less ATP.
- Diffusion = particles down the gradient. Osmosis = water. Permeability decides which. Active = uphill, costs ATP.
- Hypernatremia: water leaves, cell shrinks. Pump failure: sodium and water enter, cell swells.
- Rest: inside negative, because K+ left and anions stayed. Depolarize: Na+ in. Repolarize: permeability resets, then the pump, which needs oxygen.
- Hypoxia path: no ATP, anaerobic acid, enzyme leak (CK, LD, troponin), self-digestion, swell, calcium in, cell death.
- Adapt: atrophy, hypertrophy, hyperplasia, metaplasia, dysplasia. Then reversible injury, then necrosis.